

**PRODUCT**

MaternaLink (in development) — MVP Product Requirements Document (PRD)

Vytalix company-build documentation · reference MATERNALINK_PRD

Status labels apply throughout: KNOWN, VERIFIED, ASSUMPTION, ESTIMATE, TO VALIDATE, PROPOSED. Nothing in this document claims traction, approvals, partners or revenue that do not exist. Governing file: FACTS_BASE.

PREPARED FOR

The Founder, Vytalix

DATE

8 September 2026

VERSION

1.0 · Confidential draft

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Product: MaternaLink Connect 1.0 (internal MVP name) · **Owner:** CPO/CTO, Vytalix · **Status:** PROPOSED v0.1 (draft for discovery sign-off) · **Date:** 7 September 2026 · **Governing files:** `/FACTS_BASE.md` (language rules, labels) and `/docs/04_MATERNALINK_PRODUCT_STRATEGY.md` (strategy; this PRD implements its \$5–\$6 scope). Companion files: `/product/MATERNALINK_MVP_SPEC.md` (technical spec), `/product/MATERNALINK_ROADMAP.md` (8-quarter plan), `/docs/07_TECHNOLOGY_ARCHITECTURE.md` (group stack).

Verification note. Six web searches were used for this batch of documents, all on regulatory facts; their URLs are cited inline. Every other external figure (SMS tariffs, vendor prices, market data) is marked ESTIMATE or TO VALIDATE. Nothing here is a claim about a working product: MaternaLink is in development, the existing Vercel prototype is of unknown provenance (FACTS_BASE §1.3), and no clinical, regulatory or government status exists.

Executive view (5 lines)

1. This PRD defines the 12–16-week MVP of MaternaLink (in development) as a **non-diagnostic care-coordination, communication, education and monitoring-support platform**: enrolment, schedules and reminders, signed-off education, owned two-way messaging with human escalation, a displayed-not-interpreted diary, a care-team console, a structured handover summary, an escalation log, defaulter lists, a programme dashboard, admin/audit, and a Nigeria SMS channel.
2. **Non-goals are the product**: no risk score, no threshold alerts, no urgency classification, no symptom checker, no chatbot advice, no EPR write-back, no partner access. These keep the intended use outside medical-device classification (TO VALIDATE with regulatory counsel) and keep the safety case tractable.
3. 118 numbered requirements are prioritised MoSCoW; the Must set is deliberately small enough for a 3–4-person build team, with offline-first sync rules, a low-bandwidth mode and an SMS channel specified as first-class requirements, not afterthoughts.
4. Nigeria channel strategy is **SMS first (transactional route, registered sender ID, DND-aware), WhatsApp in v1.1, USSD and voice IVR in v1.1–v2**, with Africa's Talking or Termii as the primary gateway candidates and Twilio/Infobip as alternatives — all TO VALIDATE on price, coverage and compliance.
5. The MVP is accepted only when the end-to-end antenatal journey passes the acceptance criteria in §11 under a signed DCB0129 clinical safety case, a completed DPIA, a passed penetration test and an accessibility audit — none of which exist today.

1. Purpose, context and constraints

Item	Position (labelled)
What exists today	A concept deck (2025), an Ekiti proposal (March 2026, outcome unknown), a Vercel prototype dashboard of unknown code ownership, security and data handling, and a "Lang Switcher" demo video (VERIFIED as artefacts only; FACTS_BASE §1)
What this PRD assumes	The prototype is a demo asset, not a product asset; the MVP is specified as a rebuild with optional reuse of audited UI components (ASSUMPTION pending the prototype audit in strategy §25)
Intended use (PROPOSED, TO VALIDATE with counsel)	Support communication, education, scheduling and information handover between women receiving maternity care and their care teams, and support programme administration. Not intended to diagnose, prevent, monitor, predict, prognose, treat or provide clinical decision support (strategy §21.1)

Item	Position (labelled)
Regulatory posture	Designed to sit outside UK MDR 2002 device classification; qualification decision to be documented per feature; DTAC 2.0 form (in force from 6 April 2026, required before pilots as well as procurement — source: https://www.burges-salmon.com/articles/102mnjh/new-nhs-digital-technology-assessment-criteria-what-health-tech-suppliers-need-t/ and https://innovation.nhs.uk/news/updated-nhs-england-digital-technology-assessment-criteria-dtac-form-and-guidance/) completed as part of MVP exit
Markets in scope for MVP	One UK service-evaluation site (community midwifery team); one Nigeria SMS cohort with an NGO or state partner (neither secured — TO VALIDATE)
Budget and timeline	12–16 weeks build; cost range in <code>/product/MATERNALINK_MVP_SPEC.md</code> \$12 (ESTIMATE £140k–£320k depending on team model)

2. Goals and non-goals

2.1 Goals (MVP)

#	Goal	Measure at MVP exit (design target, ESTIMATE)
G1	A woman and her care team can complete an antenatal episode end-to-end in the product	100% of the acceptance scenarios in §11 pass in staging with test data
G2	Every message from a woman has a human owner and is acknowledged within an SLA, with escalation to humans when it is not	≥95% acknowledged within SLA in evaluation; zero unowned messages in audit
G3	Handover between professionals uses a structured, audited summary	≥60% of evaluation-site births have a summary generated
G4	A Nigeria programme can enrol a cohort and run scheduled two-way SMS with a defaulter list, without a smartphone on the woman's side	≥80% SMS delivery to enrolled numbers; defaulter list generated daily
G5	The product ships with its assurance artefacts	DCB0129 safety case v1 signed by CSO; DPIA complete; pen test passed (no unresolved high/critical); WCAG 2.2 AA audit; DTAC 2.0 form completed; Cyber Essentials certificate
G6	The product works on poor connections and intermittently offline	Core woman-side flows usable offline; sync conflict rate <1% of synced records

2.2 Non-goals (explicit, MVP and v1)

#	Non-goal	Why
NG1	Any risk score, prediction, or "Maternal Instability Score"	v3 research track only; requires validation and SaMD pathway (strategy §8)
NG2	Threshold alerts on self-recorded observations (BP, weight, movements)	Monitoring intended purpose → likely device; v2 qualification decision (strategy §7)
NG3	Automated urgency/priority classification of message content	Would make the system decide clinical urgency; hazard H-01
NG4	Symptom checker, chatbot, generated clinical advice, LLM output shown to women	Content is pre-approved only (strategy §16)
NG5	Scored mental-health questionnaires (EPDS, Whooley, GAD-2)	v2 under clinical governance review

#	Non-goal	Why
NG6	EPR/FHIR write-back, NHS login, PDS lookup	v2; MVP provides pasteable/printable outputs
NG7	Partner/companion accounts	Safeguarding design first; v2
NG8	Video consultations, wearables, device pairing, payments	Out of MVP
NG9	Being the legal record of care	The EPR/facility register remains the record; MaternaLink holds communication and coordination data
NG10	Native iOS/Android apps	PWA only in MVP; store apps considered after evaluation

3. Users and roles

Roles are defined in strategy §3 and the permission matrix is in the MVP spec §6. Summary:

Role key	Who	Primary MVP surface
<code>mother</code>	Enrolled woman (UK: smartphone PWA; Nigeria: SMS, optionally PWA)	Home, schedule, learn, diary, messages, my team, settings
<code>midwife</code> (variant <code>care_worker</code> with reduced permissions)	Community midwife; CHEW/CHIPS agent	Caseload, inbox, contact log, handover, escalation, defaulter list
<code>clinician</code>	Obstetrician, facility doctor/nurse-midwife	Caseload (facility scope), handover receipt, escalation, care notes
<code>programme_manager</code>	Head of midwifery, digital midwife, MoH desk officer, NGO M&E lead	Dashboard, exports, rotas, content library view
<code>operator_admin</code>	Trust IT/IG; state IT; implementing partner admin	Organisation, sites, users, roles, consent texts, content publishing, audit viewer
<code>vytalix_support</code>	Vytalix support/clinical safety/data protection	Break-glass, tenant configuration, incident tooling — all audited and time-boxed

Persona references: P1 Amara (UK mother), P2 Bolanle (Nigeria mother, feature phone), P3 Sarah (community midwife), P4 Mr Okafor (clinical lead), P5 Dr Adeyemi (state director), P6 Grace (NGO programme officer) — strategy §3.2 (ASSUMPTION until discovery interviews).

4. Scope map (features → requirement groups)

```

flowchart LR
    subgraph Woman
        F1[F1 Enrolment & consent] --> F3[F3 Schedule & reminders]
        F3 --> F4[F4 Education]
        F4 --> F6[F6 Diary - displayed only]
        F6 --> F5[F5 Messaging]
    end
    subgraph CareTeam[Care team]
        F7[F7 Console] --> F8[F8 Handover summary]
        F7 --> F9[F9 Escalation log]
        F7 --> F10[F10 Defaulter lists]
    end
    subgraph Programme
        F11[F11 Dashboard & exports]
        F12[F12 Admin, RBAC, audit]
    end
    F5 --> F7
    F13[F13 Channels: PWA, SMS, email] --- F3
    F13 --- F5
    F14[F14 Safeguarding behaviours] --- F1
    F14 --- F5
    F11 --- F7
    F12 --- F7

```

5. Functional requirements (numbered, MoSCoW)

Legend: **M** Must (MVP cannot ship without), **S** Should (in MVP if capacity allows; otherwise v1.1), **C** Could (v1.1/v1.2), **W** Won't (this release; recorded to prevent scope creep). "Woman" is used for the **mother** role. Every requirement that touches personal data is subject to the DPIA and every safety-relevant one is cross-referenced to a hazard ID (H-nn) in the DCB0129 hazard log (to be opened; IDs here are provisional).

5.1 F1 — Enrolment, consent, preferences

ID	Requirement	MoSCoW	Notes / hazard
FR-001	A staff user (midwife , clinician , operator_admin) can enrol a woman with minimum fields: name, date of birth, phone number, preferred language, preferred channel, organisation/site, expected date of delivery (EDD) or last menstrual period (LMP) with EDD computed and editable, and consent record	M	H-05 (wrong woman): duplicate check on phone + DOB
FR-002	A woman can self-enrol via an organisation-specific invitation link or SMS keyword, subject to staff confirmation before she is added to a caseload	S	Nigeria cohorts may self-register by keyword
FR-003	Consent is layered and granular: (a) platform use, (b) SMS/voice contact, (c) content of SMS (neutral vs. pregnancy-specific), (d) research contact, (e) programme reporting (aggregate). Each is timestamped, versioned to the consent text version, and withdrawable	M	Consent texts per organisation and language; UK GDPR / NDPA
FR-004	Withdrawal of consent is possible in-app and by SMS keyword (e.g., "STOP") and takes effect within one sync cycle; staff are notified; data handling follows the retention rule in the DPIA	M	NCC DND-style opt-out behaviour mirrored in product

ID	Requirement	MoSCoW	Notes / hazard
FR-005	Age at enrolment is captured; under-18 enrolment triggers the organisation's configured safeguarding workflow (notify safeguarding lead; require staff attestation of local consent policy)	M	Strategy §19
FR-006	Interpreter/advocate need is recorded as a flag with free text (e.g., "Yoruba, husband must not be interpreter") visible to all care-team roles	M	Handover field
FR-007	A woman can choose reminder time windows (e.g., 08:00–20:00) and quiet days	S	Feature-phone battery pattern (P2)
FR-008	Enrolment works offline for staff (queued, synced later) with a provisional ID and duplicate resolution on sync	S	Sync rules §6.3
FR-009	Bulk enrolment by CSV upload (programme cohort) with validation report	S	Nigeria NGO cohorts
FR-010	Companion/partner enrolment	W	v2

5.2 F2 — Pregnancy profile

ID	Requirement	MoSCoW	Notes / hazard
FR-011	Profile holds: EDD (with source: LMP/scan/manual), parity (gravida/para), care team assignment (named midwife/CHEW, facility), planned place of birth, birth preferences (free text and structured options), language/interpreter, allergies and known conditions as entered by clinical staff only , and "things I want my team to know" (woman-entered free text)	M	Clinical fields are staff-only; labelled "as recorded by [name] on [date]"
FR-012	EDD change re-generates the schedule and shows a confirmation with what will change; superseded reminders are cancelled	M	H-03 (stale schedule)
FR-013	Birth outcome recording (date, place, mode as free-text/coded, live birth/loss) switches the episode to postnatal and changes content and schedule; loss triggers a bereavement content pathway and suppresses all pregnancy-progress content within one sync cycle	M	H-07 (content after loss) — highest-severity content hazard
FR-014	Every profile change is audited (who, when, old→new)	M	Audit §MVP spec §7
FR-015	Multiple concurrent pregnancies per woman are not supported; a new episode can be opened after closure	M	Simplifies data model

5.3 F3 — Contact schedule and reminders

ID	Requirement	MoSCoW	Notes / hazard
FR-016	Schedule templates are configurable per organisation: UK default aligned to the NICE NG201 antenatal contact schedule; Nigeria default aligned to the WHO 8-contact model (both TO VALIDATE against current guideline text by the CSO before use)	M	Templates are clinically signed off (content governance §9)
FR-017	Schedule generation from EDD/LMP and parity produces dated contacts with type (booking, scan, ANC contact, postnatal visit, custom) and status (planned, attended, missed, cancelled, rescheduled)	M	
FR-018	Reminders are sent per contact at configurable offsets (e.g., 7 days, 1 day, morning of) via the woman's chosen channel, within her time window	M	Channel rules §6.5
FR-019	Reminder content is neutral by default on SMS ("You have an appointment on Tue 14 Oct at Ikere PHC") unless the woman opted into pregnancy-specific wording	M	H-09 (shared phone disclosure)

ID	Requirement	MoSCoW	Notes / hazard
FR-020	Staff can add, move, cancel and mark contacts attended/missed from the console; woman can request a reschedule (message type)	M	
FR-021	Missed contact (no attendance recorded by end of day + grace period) is flagged to the defaulter list (F10)	M	
FR-022	Two-way SMS confirmation ("Reply 1 to confirm, 2 to reschedule")	S	Gateway must support two-way (\$6.5)
FR-023	Calendar (.ics) export for the woman	C	
FR-024	Automatic slot booking against facility diaries	W	Requires integration

5.4 F4 — Education library

ID	Requirement	MoSCoW	Notes / hazard
FR-025	Content items have: title, body (text), optional audio, optional short video (low-data variant mandatory), language, literacy level tag, gestation window, market, sources cited, author, clinical reviewer, version, review-by date, status (draft/in review/approved/retired)	M	Content governance \$9
FR-026	Content is served to the woman by gestation window, language and market; a "what to expect this week" surface is the home screen default	M	
FR-027	Initial approved library at MVP exit: ~40 items UK English; ~20 items Nigeria English; Yoruba subset (≥10 items) if a Yoruba-speaking pilot is confirmed	M	ESTIMATE volumes; sources NICE/RCOG/WHO/FMOH patient materials (licensing TO VALIDATE)
FR-028	A permanent, non-dismissable emergency banner on every woman-facing screen with the organisation-configured emergency instruction and number (UK: "call 999 or go to your maternity unit"; Nigeria: configured facility/ambulance number)	M	H-01
FR-029	Danger-sign education is delivered as generic education ("if you have a severe headache...contact your team now") — never personalised interpretation of a woman's own recorded values	M	Intended use
FR-030	Content can be pushed as an SMS digest (≤160 chars, or concatenated up to 3 segments) for feature-phone users, with a keyword to request more	M	Nigeria
FR-031	Content engagement (opened, completed, audio played) is recorded per item per woman for the dashboard and for the woman's own progress	S	Analytics \$10
FR-032	Retiring an item removes it from all surfaces within one sync cycle and records a content-incident if retirement was due to error	M	H-06 (content error)
FR-033	Offline availability: approved text content for the woman's current and next gestation window is cached on the device; audio/video cached only on Wi-Fi or explicit download	M	Low-bandwidth \$6.4

5.5 F5 — Secure asynchronous messaging

ID	Requirement	MoSCoW	Notes / hazard
FR-034	A woman can send a message (text; photo attachment S) to her care team; each message is a thread item with a type chosen by the woman: question, appointment, "I want to tell my team something", "important"	M	Types are operational, not clinical

ID	Requirement	MoSCoW	Notes / hazard
FR-035	Every incoming message has an owner (the named midwife/CHEW; else the team inbox) and an acknowledgement SLA configured per organisation (default 4 working hours; TO VALIDATE with the evaluation site)	M	H-01 (unread message)
FR-036	If not acknowledged within SLA, the message is auto-reassigned to the on-call/lead per the rota and the programme manager is notified; a second breach escalates again; all steps audited	M	H-01
FR-037	Acknowledgement is a distinct action from reply ("Seen by Sarah at 10:12"), shown to the woman	M	Ockenden "listening" evidence
FR-038	Staff replies can use organisation-approved templates (e.g., "Thank you, I have read this; we will discuss at your visit on...")	S	
FR-039	Sending a message marked "important" or containing the woman-chosen "I am not safe" type shows the emergency banner and the care-team phone number immediately, and routes to the on-call human regardless of hours	M	H-01, H-10
FR-040	The system never classifies message urgency automatically	M	NG3
FR-041	Messages are stored end-to-end within the tenant boundary, encrypted at rest; there is no e-mail relay of message content; notifications say "You have a new message" only	M	H-09
FR-042	SMS two-way: a woman's SMS reply is ingested into her thread; staff replies out of the console are delivered by SMS when SMS is her channel; SMS previews are neutral	M	Gateway inbound webhook
FR-043	Out-of-hours behaviour: an auto-reply states the service hours and the emergency instruction; the message still enters the inbox and the SLA clock follows configured service hours	M	Explicit rota rules
FR-044	Read receipts, edit history, deletion policy: messages cannot be deleted by users; staff can mark as "sent in error" with reason (audited)	M	Record integrity
FR-045	Message attachments (images) with size limit and client-side compression	S	Low-bandwidth
FR-046	Voice-note messages	C	Nigeria literacy; needs storage and transcription policy
FR-047	Real-time chat / typing indicators	W	Asynchronous by design

5.6 F6 — Self-recorded diary and observations (display only)

ID	Requirement	MoSCoW	Notes / hazard
FR-048	A woman can record: how I feel today (5-point scale + free text), symptoms (checklist, free text), fetal movements (free text/"as usual, less than usual"), blood pressure (if she has a cuff; two integers), weight, and "questions for my next visit"	M	
FR-049	Recorded values are displayed to the woman and her care team as entered, with the timestamp and the words "recorded by you/by the woman" — no colour coding by value, no thresholds, no trend interpretation, no alerts	M	H-08 (misinterpretation): explicit UI copy "Your care team will review this at your next contact; if you are worried now, [emergency instruction]"

ID	Requirement	MoSCoW	Notes / hazard
FR-050	Chronological list and simple non-interpretive charts (plain line over time without reference bands)	S	Reference bands would imply interpretation
FR-051	Diary entries are visible on the care-team timeline and in the pre-visit view	M	
FR-052	Diary entries can be recorded offline and queue for sync; ordering by client timestamp with server receipt time shown	M	Sync \$6.3
FR-053	Entries via SMS keyword ("BP 120 80") for Nigeria	C	Parsing risk; needs confirmation echo
FR-054	Validated questionnaires with scoring	W	v2 governance

5.7 F7 — Care-team console

ID	Requirement	MoSCoW	Notes / hazard
FR-055	Caseload view scoped to the user's team/site: name, gestation/postnatal day, next contact, last contact, unread messages, missed contacts, flags (interpreter, safeguarding-present-only, under-18)	M	Safeguarding flag shows presence, not content
FR-056	Woman timeline: contacts, messages, diary, notes, escalations, handover events in one chronological view	M	
FR-057	Team inbox with filters (mine, team, unacknowledged, breached SLA) and bulk acknowledge disabled (each acknowledgement is individual)	M	H-01
FR-058	Contact log: record a contact (date, type, mode, attended, summary free text, actions)	M	Communication note, not the legal record
FR-059	Care notes (free text, versioned; "as recorded by") with copy-to-clipboard formatted for pasting into the EPR	M	Strategy \$11 point 3
FR-060	Rota: define team members, on-call periods and a lead; used by SLA escalation	M	
FR-061	Offline read of the last-synced caseload and timelines for the staff PWA; offline write of contact logs and notes with conflict handling	S	Sync \$6.3
FR-062	Pre-visit summary card: what she recorded since last contact, unanswered questions, upcoming contacts	S	P3 need
FR-063	Assign/reassign a woman between team members with audit	M	
FR-064	EPR integration	W	v2

5.8 F8 — Structured handover summary

ID	Requirement	MoSCoW	Notes / hazard
FR-065	Generate a handover summary from recorded data: identifiers, EDD/gestation, parity, care team, planned place of birth, birth preferences, allergies/known conditions as recorded by staff (with attribution), interpreter need, safeguarding flag presence, companion, woman's "important" messages (last 30 days), open escalations, upcoming contacts	M	H-05 (wrong woman): summary carries three identifiers and a visible warning to confirm identity

ID	Requirement	MoSCoW	Notes / hazard
FR-066	Summary is editable by the sending clinician before issue; issued versions are immutable and numbered	M	
FR-067	Output: on-screen, print/PDF, copy-as-text; "acknowledge receipt" action by a receiving user, or a printable acknowledgement line when the receiving facility is not on the platform	M	Audit of view/acknowledge
FR-068	Summary contains no computed assessment or score ; a footer states it is a communication summary and the receiving clinician performs their own assessment	M	Intended use
FR-069	Shift/on-call handover list: all women with open items for the on-call period	S	
FR-070	FHIR export	W	v2

5.9 F9 — Escalation log

ID	Requirement	MoSCoW	Notes / hazard
FR-071	Staff can record an escalation once: to whom (role/name), by what means (phone, in person, referral), when, reason (free text), outcome, follow-up owner	M	Ockenden escalation actions
FR-072	Open escalations appear on the caseload, timeline and handover summary until closed	M	
FR-073	Programme managers see aggregate counts and time-to-close; individual content is visible only to care roles	M	Privacy
FR-074	Referral to another facility (Nigeria) with a receiving-facility notification by SMS to a configured facility number	S	P2 success condition

5.10 F10 — Defaulter / missed-contact tracing

ID	Requirement	MoSCoW	Notes / hazard
FR-075	Daily list per team/site of women with missed contacts or no contact in N days (configurable), each assignable to a human with status (to contact, contacted, unreachable, rebooked)	M	Operational; assigned to a human
FR-076	CHEW/field mode: list sorted by locality, phone-call button, offline-capable with status sync	S	Nigeria
FR-077	Export defaulter list (CSV) with organisation-configured field set	M	

5.11 F11 — Programme dashboard and exports

ID	Requirement	MoSCoW	Notes / hazard
FR-078	Metrics: enrolments (by site, week), active women, contact completion vs schedule, missed and traced, messages and SLA performance, escalations opened/closed, content reach, SMS sent/delivered/failed, consent withdrawals	M	Operational only; no clinical outcomes
FR-079	Small-cell suppression: no cell below 5 displayed or exported in aggregate views	M	Strategy \$15

ID	Requirement	MoSCoW	Notes / hazard
FR-080	CSV export of aggregate indicators with date range; row-level export restricted to <code>operator_admin</code> under DPIA-defined purpose and audited	M	
FR-081	DHIS2-compatible aggregate export (data element mapping configurable)	C	v1.1/1.2
FR-082	Clinical outcome reporting (mortality, morbidity)	W	Never as product metrics without a formal evaluation

5.12 F12 — Administration, RBAC, audit

ID	Requirement	MoSCoW	Notes / hazard
FR-083	Multi-tenant: organisation → sites → teams; all data scoped to organisation; no cross-tenant queries	M	Isolation control
FR-084	User management: invite, role assignment, deactivation, MFA enforcement for staff roles, session policies	M	
FR-085	Consent-text and schedule-template management with versioning and clinical sign-off state	M	
FR-086	Content publishing workflow (draft → review → approved → retired) with two-person rule	M	\$9
FR-087	Audit log viewer: filter by user, woman, action, date; export for IG; immutable	M	
FR-088	Break-glass access for <code>vytalix_support</code> : reason required, time-boxed, tenant notified, audited	M	
FR-089	Data subject request tooling: export a woman's data; rectify; delete/anonymise subject to retention rules	S	UK GDPR rights
FR-090	Organisation-level configuration: emergency instruction and numbers, service hours, SLA, channel availability, languages, market defaults	M	

5.13 F13 — Channels

ID	Requirement	MoSCoW	Notes / hazard
FR-091	Woman-facing PWA (installable, offline-capable) for UK and smartphone users in Nigeria	M	
FR-092	Staff PWA (console) with offline read	M	
FR-093	SMS outbound (reminders, digests, notifications) and inbound (replies, keywords) via one gateway abstraction with provider adapters (Africa's Talking first; Termii second; Twilio for UK)	M	\$6.5
FR-094	UK notifications: e-mail and SMS "you have a new message/appointment" with no sensitive content	M	
FR-095	Web push notifications for PWA	S	iOS support constraints TO VALIDATE

ID	Requirement	MoSCoW	Notes / hazard
FR-096	WhatsApp Business Platform channel (template messages, two-way)	C	v1.1; BSP selection \$6.5
FR-097	USSD menus (schedule, confirm, request call-back)	C	v1.1/v2
FR-098	Voice IVR reminders in local languages	C	v1.1; TTS quality for Yoruba/Hausa TO VALIDATE
FR-099	Native app-store apps	W	Post-evaluation

5.14 F14 — Safeguarding behaviours

ID	Requirement	MoSCoW	Notes / hazard
FR-100	Discreet mode: neutral app name/icon option, no sensitive content in notification previews, quick-exit button to a neutral page, PIN/biometric lock	M	H-09; Domestic Abuse Act 2021 context (strategy \$19)
FR-101	"I am not safe" message type routes to a named human with immediate notification and emergency banner (FR-039)	M	
FR-102	Staff prompt at each recorded contact to confirm routine enquiry per local policy (attestation only; no content stored unless the staff member records a concern)	S	
FR-103	Safeguarding concern record visible only to care roles and the configured safeguarding lead; presence-only flag elsewhere	M	
FR-104	No partner access; consent for any third party to receive SMS is explicit	M	

5.15 Cross-cutting

ID	Requirement	MoSCoW	Notes
FR-105	Every woman-facing screen states, in plain language, what the product is and is not ("This app helps you and your care team stay in touch. It does not check your health or tell you if something is wrong.")	M	Intended use in the UI
FR-106	Woman can see who has viewed her record (transparency log)	S	Trust
FR-107	Identity confirmation pattern (name + DOB shown) before any staff action on a woman's record after search	M	H-05
FR-108	All dates show timezone-safe values (Europe/London; Africa/Lagos)	M	
FR-109	Feature flags per organisation for every S/C item	M	Controlled rollout
FR-110	Data export on tenant exit (full organisation data in documented format)	S	No lock-in (strategy \$12)

6. Non-functional requirements

6.1 Availability, performance, capacity

ID	Requirement	Target (design target, ESTIMATE — not a contractual or marketing claim)
NFR-001	Availability of the API and web app (monthly, excluding announced maintenance)	99.5% design target for MVP; 99.9% as the v2 architecture target once multi-AZ database and blue/green deploys are proven. The Ekiti email's "99.5–99.9%" is treated as an aspiration, not a commitment (FACTS_BASE §1.2)
NFR-002	Reminder dispatch punctuality	≥99% of reminders handed to the gateway within 5 minutes of scheduled time
NFR-003	API latency	p95 < 500 ms for reads, < 1 s for writes at MVP load
NFR-004	Woman PWA first meaningful paint on 3G-class connection	< 5 s on first visit; < 2 s on repeat (cached shell)
NFR-005	MVP capacity	5 organisations, 20 sites, 200 staff, 20,000 enrolled women, 2 million SMS/year without architectural change (ESTIMATE)
NFR-006	Maintenance windows	Announced 72 h ahead; outside UK and Nigeria service hours; emergency banner and phone numbers remain reachable via static fallback page
NFR-007	Degraded mode	If the API is down, the woman PWA still shows cached schedule, content and the emergency banner; the SMS gateway queue drains when service returns

6.2 Data, privacy, security (summary — detail in MVP spec §7–§9)

ID	Requirement
NFR-010	UK data at rest in AWS eu-west-2 (London) (or Azure UK South as alternative); Nigeria deployment cell configurable per contract (strategy §15)
NFR-011	Encryption in transit TLS 1.2+; at rest AES-256 with KMS; field-level encryption for direct identifiers (name, phone, DOB)
NFR-012	Audit log of every read and write of personal data, immutable, retained ≥ 6 years (TO VALIDATE against controller retention schedules)
NFR-013	RBAC with least privilege; MFA for all staff roles; SSO (OIDC/SAML) for organisations that require it
NFR-014	No patient data leaves the tenant boundary to any third-party model endpoint in MVP (strategy §16); analytics events contain no free text or direct identifiers (§10)
NFR-015	DPIA completed and approved before any real data; DPA signed with each controller

6.3 Offline-first and synchronisation rules

Principles: the server is the system of record; clients are caches with an outbox; every write is idempotent; conflicts are resolved by explicit, documented rules, never silently.

Rule	Specification
SYNC-01 Scope of offline	Woman PWA: read schedule, content (current+next window), own diary, own messages; write diary entries, messages, reschedule requests, consent withdrawal. Staff PWA: read last-synced caseload and timelines (S); write contact logs, notes, defaulter statuses, enrolments (S)
SYNC-02 Storage	IndexedDB via a local database layer; encrypted at rest on device using a key derived from the user's PIN/biometric unlock (web-crypto); auto-purge of cached records for women no longer on the caseload; maximum 50 MB text cache per staff device (ESTIMATE)

Rule	Specification
SYNC-03 Outbox	Every offline write gets a client-generated UUID v7 and client timestamp; queued in an outbox; sent in order; server accepts idempotently (duplicate UUID = no-op with original result)
SYNC-04 Append-only entities	Messages, diary entries, escalation entries, contact-log entries, audit events are append-only ; no conflicts possible — the server orders by receipt time and displays client time alongside
SYNC-05 Versioned entities	Profile, care notes, birth preferences, schedule contacts carry a version number. A write with a stale version is rejected; the client shows both versions and the user chooses (three-way merge for free text is offered; no automatic overwrite). Hazard H-04 (sync overwrite)
SYNC-06 Server-authoritative entities	Schedule generation, rota, SLA timers, consent state, content publication state — clients never compute these; they display the last-synced value with a "last updated" stamp
SYNC-07 Consent withdrawal	Has priority in the outbox and is processed before any other queued write from that user
SYNC-08 Sync triggers	On reconnect, on app foreground, every 15 minutes when online, and on explicit user pull
SYNC-09 Clock skew	Client timestamps are displayed but server receipt time is the ordering key; skew > 10 minutes is flagged in the record
SYNC-10 Failure visibility	The user always sees the count of unsent items and can retry; unsent items older than 24 h show a persistent warning and, for messages, instruct the woman to phone her team (H-01)
SYNC-11 Staff device loss	Remote revocation of the device's refresh token; local cache is unreadable without the PIN-derived key; cache expires after 7 days without sync

6.4 Low-bandwidth mode

Rule	Specification
LB-01	"Lite" mode toggled automatically on 2G/3G/Save-Data header or manually: text-only content, no images above 20 KB, no autoplay, audio/video as explicit downloads with size shown
LB-02	Page payload budgets: app shell ≤ 300 KB compressed on first load; per-route JSON ≤ 50 KB; images served via responsive variants (WebP/AVIF)
LB-03	Delta sync: only records changed since the last sync cursor
LB-04	Content bundles for offline: per gestation window, text-only, ≤ 200 KB
LB-05	SMS fallback: any woman on PWA can also enable SMS reminders so that she is reachable when data is off
LB-06	Compression (Brotli/gzip) and HTTP/2 or HTTP/3 through the CDN

6.5 Channel strategy — Nigeria (SMS, USSD, WhatsApp, voice)

Principle: meet the woman where she is. Feature phones dominate outside cities (ASSUMPTION; NDHS/GSMA figures TO VALIDATE); therefore SMS is the Must channel, with WhatsApp as the smartphone upgrade path and USSD/voice for interactive and low-literacy use.

Channel	MVP status	Use	Constraints (regulatory facts cited; commercial figures TO VALIDATE)
SMS (two-way)	M	Reminders, digests, replies, keywords (CONFIRM, STOP, HELP, MORE)	Sender ID must be registered with the NCC process via the operators (MTN, Glo, Airtel, 9mobile); alphanumeric IDs up to 11 characters; the Do-Not-Disturb service on short code 2442 blocks promotional traffic to opted-out subscribers, while transactional/informational messages can still be delivered via corporate transactional routes. Sources: https://www.sent.dm/en/resources/sms-compliance/nigeria-sms-guide ; https://arkesel.com/bulk-sms-in-nigeria-legal-and-regulatory-aspects/ ; https://www.bulksmsnigeria.com/resources/sms-compliance-nigeria . MaternaLink reminders are informational/transactional by design (opt-in, service messages) — classification with the gateway and counsel TO VALIDATE. Per-SMS cost ESTIMATE ₦4–₦6 (strategy \$9) TO VALIDATE
WhatsApp Business Platform	C (v1.1)	Rich reminders, content cards, two-way chat for smartphone users	Requires a Business Solution Provider or direct Cloud API access; template pre-approval; conversation-based pricing (TO VALIDATE current rates); data flows via Meta — DPIA and NDPA cross-border analysis required before enabling
USSD	C (v1.1/v2)	Menu: next appointment, confirm, request call-back, danger-sign guide	Short-code lease via an aggregator; session timeouts (~180 s); costs per session TO VALIDATE; works on any phone with no data
Voice IVR / voice reminders	C (v1.1)	Yoruba/Hausa/Igbo/Pidgin recorded reminders; call-back requests	Pre-recorded human audio preferred over TTS for local languages (quality TO VALIDATE); per-minute cost TO VALIDATE
PWA push	S	Smartphone users	Requires installation and permission; iOS constraints TO VALIDATE

Provider options (all TO VALIDATE on pricing, delivery rates, DND handling, data residency and contract terms):

Provider	Channels	Strengths	Concerns	Role
Africa's Talking	SMS, USSD, voice, airtime	Africa-native, USSD and voice in one API, sandbox, Nigeria presence	Enterprise support, SLA terms TO VALIDATE	Primary candidate (Nigeria)
Termii	SMS, WhatsApp, voice, OTP	Nigeria-native, DND-aware routes, WhatsApp	Multi-country reach, enterprise SLA TO VALIDATE	Secondary / failover (Nigeria)
Twilio	SMS, WhatsApp (BSP), voice	Global, mature tooling, UK SMS	Cost per SMS in Nigeria often higher; DND/sender-ID handling via local partners TO VALIDATE	UK SMS ; global fallback
Infobip	SMS, WhatsApp, USSD, voice	Enterprise, local Nigerian connectivity, WhatsApp BSP	Contract minimums TO VALIDATE	Enterprise alternative
Meta WhatsApp Business Platform (direct Cloud API)	WhatsApp	No BSP margin; template control	Business verification; conversation pricing; support	v1.1 option via BSP first

Architecture requirement: a channel abstraction (`ChannelProvider` interface: send, receive-webhook, delivery-status, opt-out sync) so that providers are configuration, not code changes; per-organisation provider selection; delivery status stored per message; cost recorded per message for pass-through billing.

Compliance requirements: opt-in recorded before first SMS; STOP handled in all languages; DND list respected for anything not clearly transactional; sender ID registered before pilot; message logs retained per DPIA; no clinical content in SMS beyond generic education; Nigeria Data Protection Act 2023 applies — registration with the NDPC as a data controller/processor of major importance is likely required because health is a designated sector and the threshold is reportedly >200 data subjects in 6 months (TO VALIDATE against the current NDPC General Application and Implementation Directive: <https://ndpc.gov.ng/wp-content/uploads/2025/07/NDP-ACT-GAID-2025-MARCH-20TH.pdf> ; secondary source: <https://globaladvisoryexperts.com/ndpc-data-controller-registration/>).

6.6 UK notification channel

E-mail (transactional provider) and SMS (Twilio or UK-native alternative such as a GOV.UK Notify-style service if eligible — TO VALIDATE eligibility) carry **no personal or clinical content**; only "You have a new message in MaternaLink" with a deep link.

7. Accessibility

ID	Requirement
ACC-01	WCAG 2.2 AA conformance for both PWAs; audited by an external assessor before go-live (ESTIMATE £3k–£6k)
ACC-02	Alignment with the NHS service standard and NHS design system patterns for UK deployments (component reuse where licence allows; TO VALIDATE)
ACC-03	Reading age: woman-facing English content at reading age 9–11 (plain English); tested with a readability tool and user testing
ACC-04	Audio versions of all Must content items in each supported language
ACC-05	Large-touch targets (≥ 44 px), high-contrast theme, dynamic text scaling, screen-reader labels on all controls, no colour-only meaning
ACC-06	Feature-phone parity: every Must woman-side function has an SMS equivalent (reminder, confirm, message, STOP)
ACC-07	Works on low-end Android (2 GB RAM, Android 9+) and iOS 15+ Safari (TO VALIDATE PWA constraints); Chrome, Edge, Firefox, Safari last two versions on desktop for staff
ACC-08	Discreet mode and quick exit are keyboard and screen-reader accessible

8. Localisation

Item	Position
Now (MVP)	English UI for all roles; i18n framework in place (ICU message format, locale files, RTL-capable although not required); Yoruba content subset (≥10 items) if a Yoruba-speaking pilot is confirmed; SMS templates in English and Yoruba
Roadmap	Yoruba UI (v1.1), Hausa (v1.2/v2, required for northern states), Igbo (v2), Nigerian Pidgin (v1.1 for SMS/voice — high reach, low translation cost, ASSUMPTION), further UK community languages by content only (Urdu, Bengali, Polish, Somali, Arabic — priorities TO VALIDATE with the evaluation site)

Item	Position
Existing "Lang Switcher" demo	A language-switcher demo video exists among the Ekiti proposal materials (VERIFIED as a video only; FACTS_BASE \$1.2). It demonstrates UI toggling, not translated clinical content or a translation QA process. It may be referenced as a UX intention; nothing from it is reused until the prototype audit (strategy \$25) clears code ownership
Translation process	Professional translation by native speakers with maternal-health experience → back-translation → clinical reviewer sign-off in the target language → user testing with ≥5 women per language; glossary maintained; machine translation may draft internal first passes only, never published unreviewed (strategy \$16)
Locale rules	Dates, times, numbers per locale; timezones Europe/London and Africa/Lagos; names in free text (no forced first/last split); phone numbers in E.164
Cultural adaptation	Content reviewed for local practice (e.g., traditional birth attendants, family decision-making, faith contexts) with the implementing partner; pictures and audio recorded locally
Cost (ESTIMATE)	£0.10–£0.20 per word professional translation plus clinical review; ~15,000 words per language for the initial library → £2k–£4k per language plus review and audio recording £1k–£3k

9. Content governance

Element	Rule
Sources	NICE, RCOG/RCM patient information, NHS.uk, Tommy's, WHO ANC recommendations, Nigerian FMOH/NPHCDA materials — licensing and reuse rights TO VALIDATE per source
Authorship	Clinical author (midwife/obstetrician) drafts; two-person rule : second clinical reviewer or CSO approves; both recorded on the item
Versioning	Every item has version, sources, author, reviewer, approval date, review-by date (≤ 12 months); superseded versions retained
Review triggers	Guideline change, incident, user feedback, translation change, 12-month expiry
Content incident	Any error reaches the CSO within 24 h; item retired within one sync cycle (FR-032); women who received it are identified and re-messaged where clinically indicated by the deploying organisation
Claims	No content may state that MaternaLink detects, predicts or prevents any condition; all danger-sign content ends with the organisation's emergency instruction
Local overlays	Organisations can add local items (facility directions, local numbers) under the same workflow
Bereavement pathway	Dedicated content set, reviewed with a bereavement specialist; automatic suppression of pregnancy-progress content after loss (FR-013)
Language parity	An item is "approved" in a language only when that language version has passed the translation process (\$8)

10. Analytics events

Rules: events carry pseudonymous IDs (hashed woman ID per tenant), organisation and site, role, locale, channel and app version; **never** free text, phone numbers, names, DOB, message content or diary values. Event storage is in-region; product analytics run on the events table in the warehouse, not in a third-party SaaS, until a DPIA permits otherwise. Sampling is not used for safety-relevant events.

Event	Fired when	Key properties	Used for
enrolment_started / enrolment_completed / enrolment_abandoned	Enrolment flow	channel, self_vs_staff, step	G1, funnel
consent_given / consent_withdrawn	Consent change	consent_type, consent_version	Compliance reporting
schedule_generated / contact_status_changed	Schedule events	contact_type, status, days_from_due	Contact completion
reminder_sent / reminder_delivered / reminder_failed	Gateway callbacks	channel, provider, offset, failure_code	Delivery rate
content_viewed / content_completed / audio_played	Education	content_id, language, gestation_window, lite_mode	Reach
diary_entry_created	Diary	entry_type (not value), offline_flag	Engagement
message_sent (woman) / message_received (staff)	Messaging	message_type, channel, offline_flag	Engagement
message_acknowledged / sla_breached / message_reassigned	SLA engine	minutes_to_ack, breach_level	G2 (safety metric)
escalation_opened / escalation_closed	F9	hours_open	Programme
handover_generated / handover_acknowledged	F8	version, output_type	G3
defaulter_listed / defaulter_status_changed	F10	status, days_to_trace	Programme
sync_completed / sync_conflict	Sync engine	records, duration, entity_type	G6
lite_mode_enabled	Auto/manual	trigger	Low-bandwidth
discreet_mode_enabled / quick_exit_used	Safeguarding	— (no further properties)	Design review only; low-cardinality
break_glass_access	Support	reason_code	Governance
error_client / error_api	Runtime	code, route (no payload)	Quality

11. Acceptance criteria (MVP exit)

Format: Given / When / Then. All must pass in the staging environment with synthetic data, and the safety-relevant ones must be re-run in production smoke tests with test tenants.

AC	Scenario
AC-01 Antenatal end-to-end	Given a midwife enrolls a woman with EDD, When the woman opens the PWA, Then she sees her dated contacts, this week's content, the emergency banner and her named team; and the midwife sees her on the caseload within one sync
AC-02 Reminder	Given a contact in 7 days and channel SMS, When the reminder time arrives, Then an SMS with neutral wording is handed to the gateway within 5 minutes and <code>reminder_sent</code> and the delivery status are recorded

AC	Scenario
AC-03 Message SLA	Given a message is sent at 09:00 on a working day with SLA 4 h and no acknowledgement, When 13:00 passes, Then the message is reassigned to the on-call user, the programme manager is notified, <code>sla_breached</code> is recorded and the audit log shows the chain
AC-04 Not-safe message	Given a woman selects "I am not safe", When she sends it, Then the emergency banner and team phone are shown immediately, the on-call user receives a notification within 1 minute, and the event is audited
AC-05 Diary display only	Given a woman records BP 160/110, When staff view it, Then it is shown exactly as entered with the timestamp and attribution and no colour, alert, flag or interpretation appears anywhere
AC-06 Handover	Given a woman with recorded preferences, an open escalation and an "important" message, When a midwife generates a handover summary, Then it contains those items with attribution, three identifiers, the identity-check warning, the "no assessment" footer, and a receiving user can acknowledge it (audited)
AC-07 Escalation once	Given a clinician records an escalation, When saved, Then it appears in the timeline, caseload and handover summary and in the programme aggregate without content
AC-08 Defaulter	Given a contact passes its date without attendance, When the daily job runs, Then the woman appears on the defaulter list and can be assigned and stasured; CSV export matches the on-screen list
AC-09 Offline diary	Given the woman is offline, When she records two diary entries and sends one message, Then they are visible locally with "not yet sent", and after reconnection they sync exactly once (duplicate-safe) with client and server times shown
AC-10 Conflict	Given a note is edited on two devices offline, When both sync, Then the second is rejected with both versions shown to the user, and no version is lost
AC-11 EDD change	Given EDD changes by 10 days, When confirmed, Then the future schedule is regenerated, superseded reminders are cancelled (no orphan SMS is sent) and the audit shows old→new
AC-12 Loss pathway	Given a birth outcome of loss is recorded, When the woman next syncs (or within 15 minutes for SMS), Then no pregnancy-progress content or reminder is delivered and the bereavement pathway is shown
AC-13 Consent STOP	Given a woman texts STOP, When received, Then no further SMS is sent, her consent record updates, staff are notified and the audit log records it
AC-14 Tenant isolation	Given two organisations, When any user of one searches, lists or requests any record by ID from the other, Then the API returns not-found and the attempt is logged
AC-15 Discreet mode	Given discreet mode is on, When a notification arrives, Then the preview contains no sensitive words and the quick-exit control leaves no MaternaLink page in the visible history
AC-16 Accessibility	Given the external audit, When completed, Then no WCAG 2.2 AA failures remain unresolved
AC-17 Security	Given the penetration test report, When reviewed, Then no critical or high findings remain open, and the retest confirms fixes
AC-18 Assurance pack	The DCB0129 clinical safety case report v1 is signed by the CSO, the DPIA is approved by the DPO, the DTAC 2.0 form is completed, and Cyber Essentials is certified (Plus scheduled)
AC-19 Lite mode	Given a simulated 3G connection with Save-Data, When the woman opens the app, Then the shell loads under 5 s and the content route payload is under 50 KB
AC-20 Nigeria cohort	Given a programme officer uploads a CSV of 200 women with phone numbers and EDDs, When processed, Then a validation report is produced, schedules are generated, opt-in SMS are sent and replies are ingested into threads

12. Risks (PRD-level)

Risk	Likelihood	Impact	Mitigation
A Must feature is judged a medical device (e.g., diary display, handover)	Medium	High	Intended-use statement in UI; no interpretation anywhere; documented qualification decision per feature; regulatory counsel review before go-live
SLA escalation fails silently (no on-call configured)	Medium	Critical	Rota is mandatory before an organisation goes live; daily job checks rota coverage; alert to Vytalix support on gaps
SMS delivery failures or DND blocking in Nigeria	Medium	High	Transactional route, registered sender ID, delivery status tracking, failover provider, voice/USSD roadmap
Content error reaches women	Medium	High	Two-person rule, review dates, content-incident process, rapid retirement
Offline sync data loss	Low	High	Append-only entities, versioned writes, idempotency, conflict UI, tested with fault injection
Shared-phone disclosure	Medium	High	Neutral SMS default, discreet mode, per-channel consent
Scope creep from the Ekiti proposal's feature list or the prototype	High	Medium	This PRD's Won't list; change control via CPO
Discovery invalidates personas/flows	Medium	Medium	Discovery before build weeks 1–2; PRD v0.2 sign-off gate
No evaluation site by build end	Medium	High	Commercial workstream in parallel; NGO cohort as fallback

13. Open questions

#	Question	Owner	Needed by
OQ-01	Who owns the prototype code and the "Maternal Link" name; can any of it be reused?	Founder	Before build week 1
OQ-02	Which UK site (and which community team) will host the service evaluation, and what SLA and service hours do they want?	Commercial	Build week 4
OQ-03	Which Nigeria partner and state; is the pilot Yoruba-speaking (drives translation scope)?	Commercial	Build week 4
OQ-04	Does the CSO agree the diary display, handover summary and defaulter list are outside device classification as specified?	CSO + counsel	Build week 2
OQ-05	Controller/processor allocation per deployment and the resulting retention periods	DPO/counsel	DPIA v1 (week 6)
OQ-06	Can NHS design system components be used under licence, and does the site require NHS login (v2)?	CTO	Week 3
OQ-07	Gateway choice and sender ID registration lead time in Nigeria	CTO + partner	Week 6
OQ-08	Is a NAFDAC view needed on non-diagnostic software in Nigeria?	Regulatory	Before pilot
OQ-09	Which content sources can be licensed (RCOG/RCM/Tommy's) versus authored in-house?	Clinical lead	Week 4
OQ-10	Does DTAC 2.0 need to be assessed by the evaluating trust before a no-fee service evaluation (the guidance says DTAC applies before pilots)?	Commercial + CTO	Week 8

Priorities / Risks / Next actions

Priorities

1. Close OQ-01 (IP/prototype) and OQ-04 (device qualification of Must features) — they determine whether the build starts as a rebuild and whether the scope is safe.
2. Run discovery (20 women + 10 professionals per market) and re-issue this PRD as v0.2 with personas validated and SLA/service-hour defaults set by the evaluation site.
3. Freeze the Must set (FR-001–FR-110 marked M) and open the DCB0129 hazard log against it before the first sprint.
4. Select and contract the Nigeria SMS gateway and start sender-ID registration; select the UK notification provider.
5. Commission the accessibility audit, penetration test and DPIA in parallel with the build so that they land in week 14–16.

Risks — see §12; the top three are device qualification of Must features, silent SLA failure, and SMS delivery/DND in Nigeria.

Next actions

- CPO/CTO: circulate this PRD to the CSO (to be appointed) and regulatory counsel; schedule discovery; draft hazard log entries H-01–H-10.
- Founder: IP agreement and prototype audit authorisation; evaluation-site and Nigeria-partner conversations.
- Clinical lead (to recruit): source and licence the initial content library; nominate the second reviewer.
- Design: information architecture and low-fidelity flows for the six personas; content-first design for SMS.